

Great — I'll prepare a full **Final Report structure** for your *WorkSync Hub* project based on your instructions. All diagram/screenshots sections will be marked clearly for you to upload images later. Here's the full content except those.

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# Final Report: WorkSync Hub

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## a) Project Introduction

**WorkSync Hub** is an advanced productivity and collaboration platform designed to streamline the workflow and enhance productivity within teams. This system integrates task management, real-time communication, document sharing, and automation tools into a centralized workspace. The goal is to reduce dependency on multiple applications and foster seamless teamwork.

### Project Objectives:

- Enhance team productivity through a unified platform.
- Simplify task management, communication, and file handling.
- Provide automation and integration tools to save time.
- Ensure a secure and intuitive user experience.

### Key Features:

1. Task Management
  2. Real-time Chat & Messaging
  3. Video Conferencing
  4. File Sharing & Storage
  5. Calendar Integration
  6. Workflow Automation
  7. Time Tracking
  8. Customizable Dashboards
  9. Multi-platform Support
  10. Role-based Access Control
  11. Offline Mode
- 

## b) User Stories (All Sprints)

### Sprint 1: Core Functionality

- As a user, I want to sign up and log in so that I can securely access my workspace.
- As a user, I want to create tasks and assign them with deadlines and priority.
- As a user, I want to chat with my team members in real-time.
- As a user, I want to upload and download files securely.

## **Sprint 2: Advanced Collaboration**

- As a user, I want to join or host video meetings.
- As a user, I want to receive productivity insights and suggestions.
- As a user, I want to automate task reminders and routine workflows.

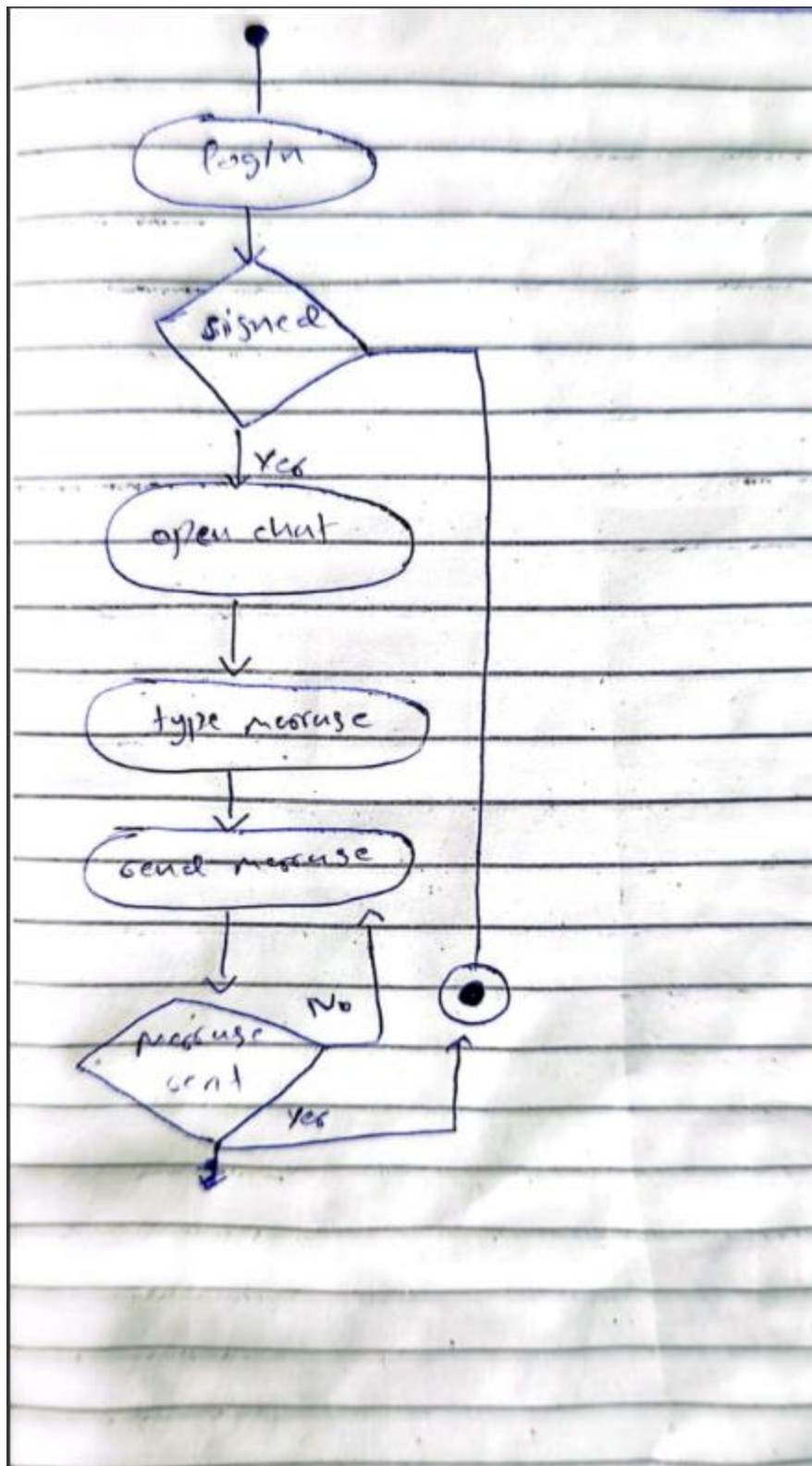
## **Sprint 3: Integration & Analytics**

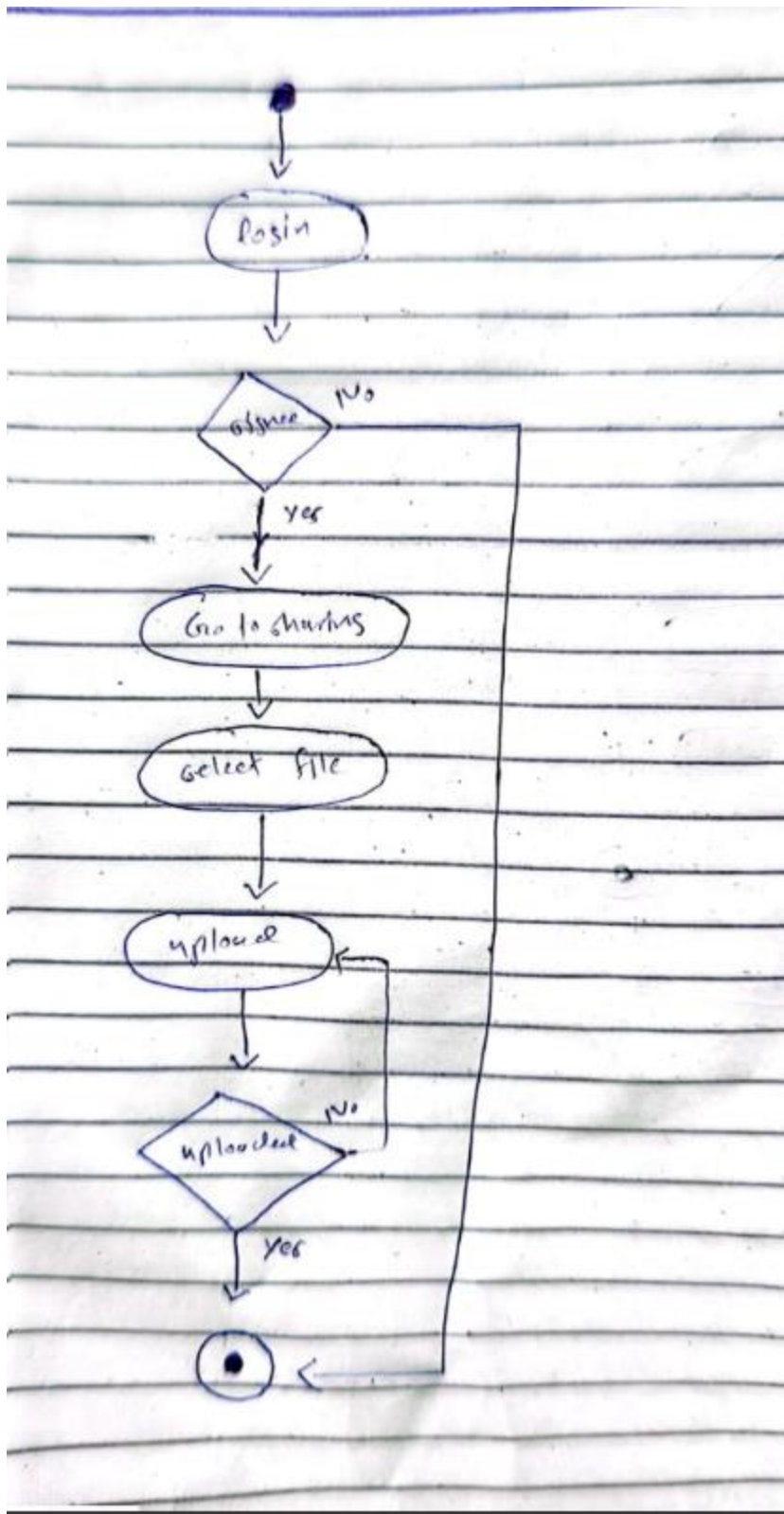
- As a user, I want to integrate my calendar with Google and Outlook.
- As a user, I want to customize my dashboard with widgets.
- As a user, I want to access data offline and sync once back online.
- As an admin, I want to control access rights based on user roles.

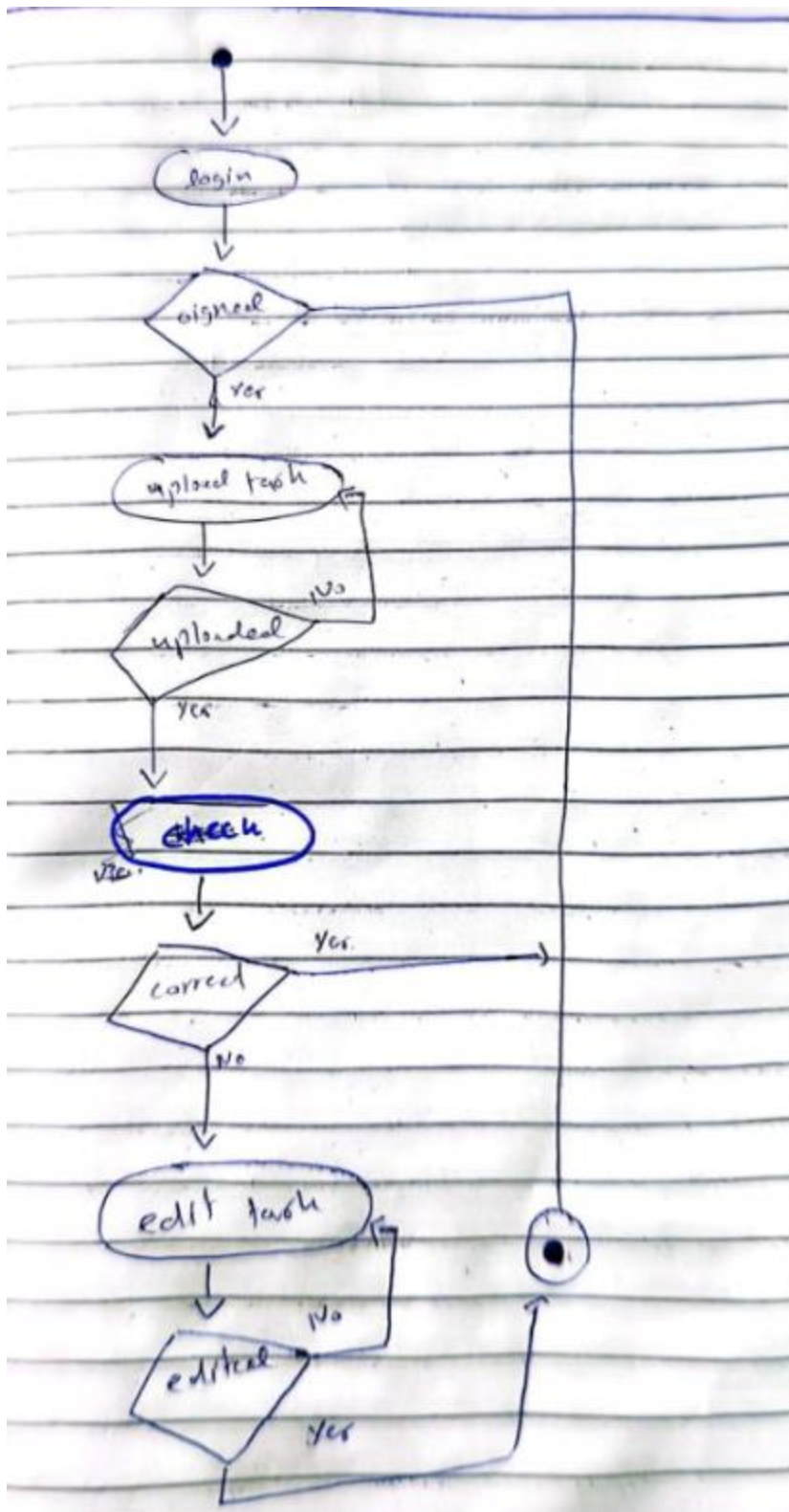
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## **c) Design (All Sprint 2 Items)**

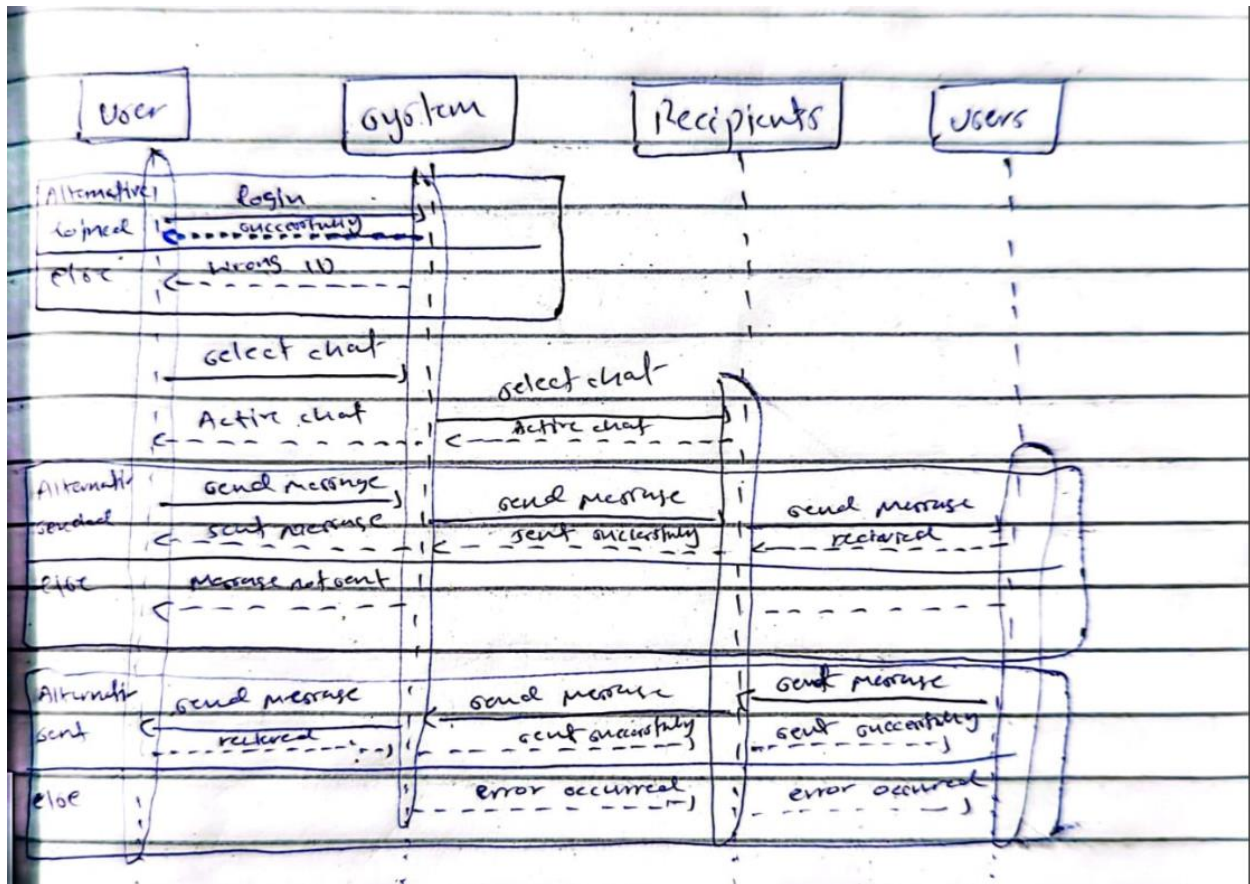
*Activity Diagrams for Task Assignment, Chat, and File Sharing*

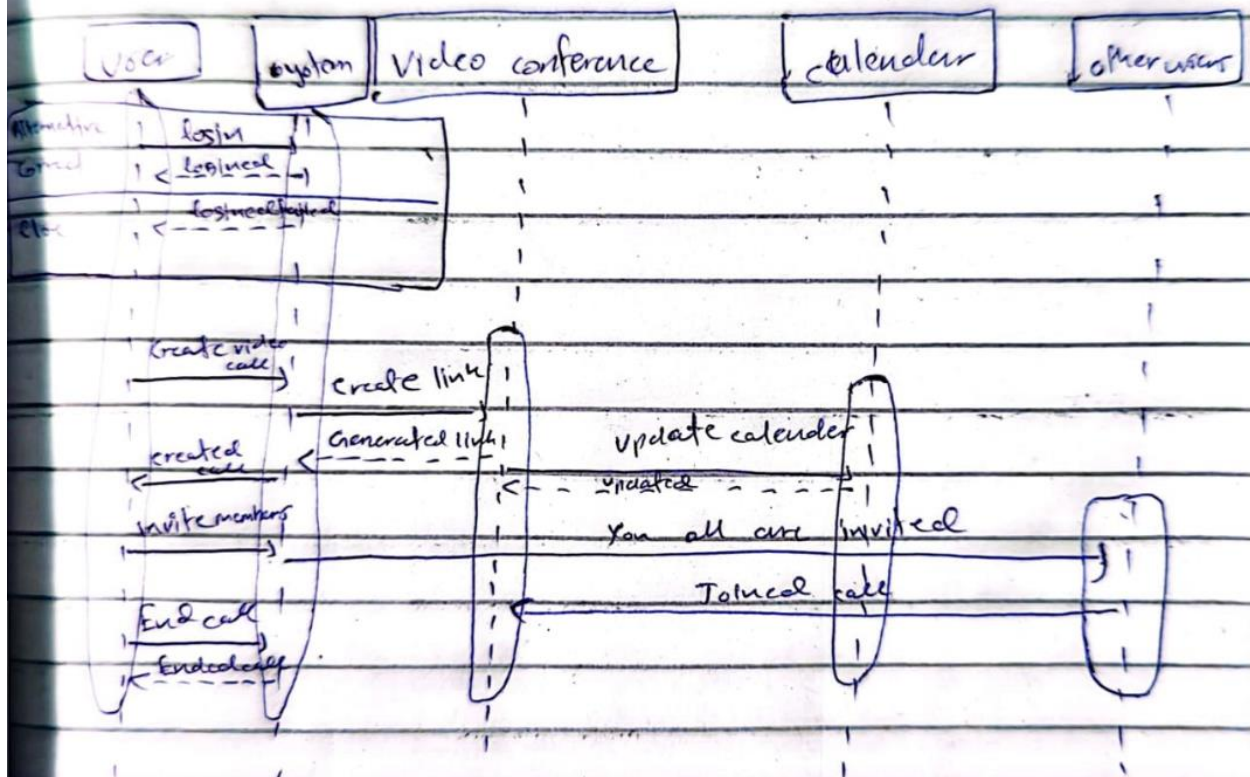


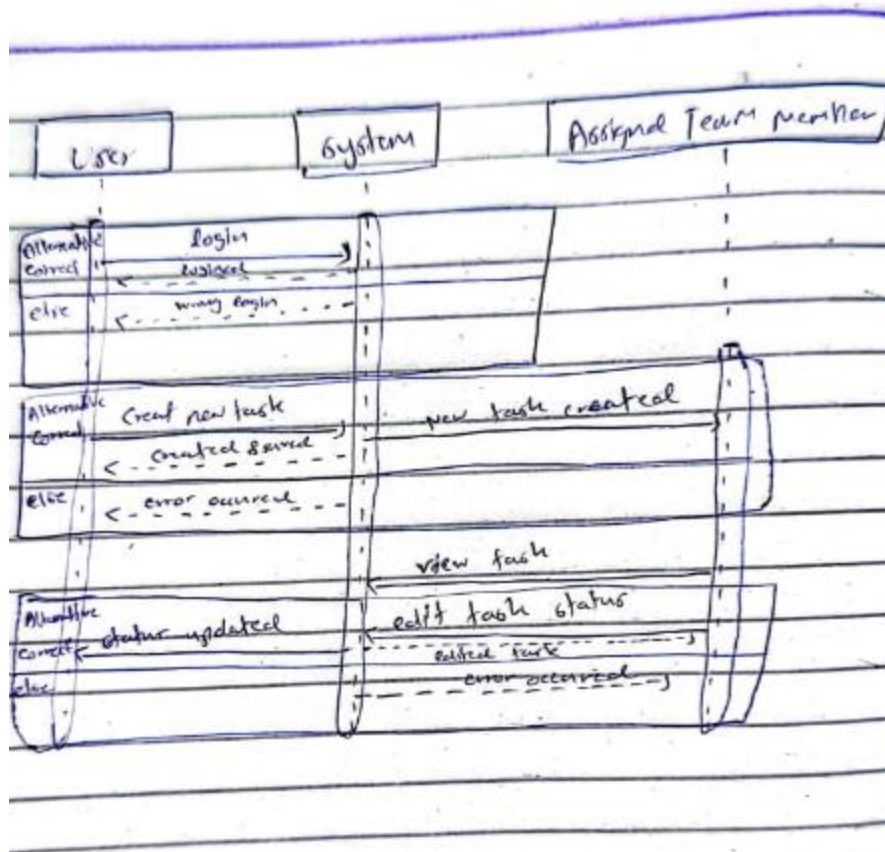




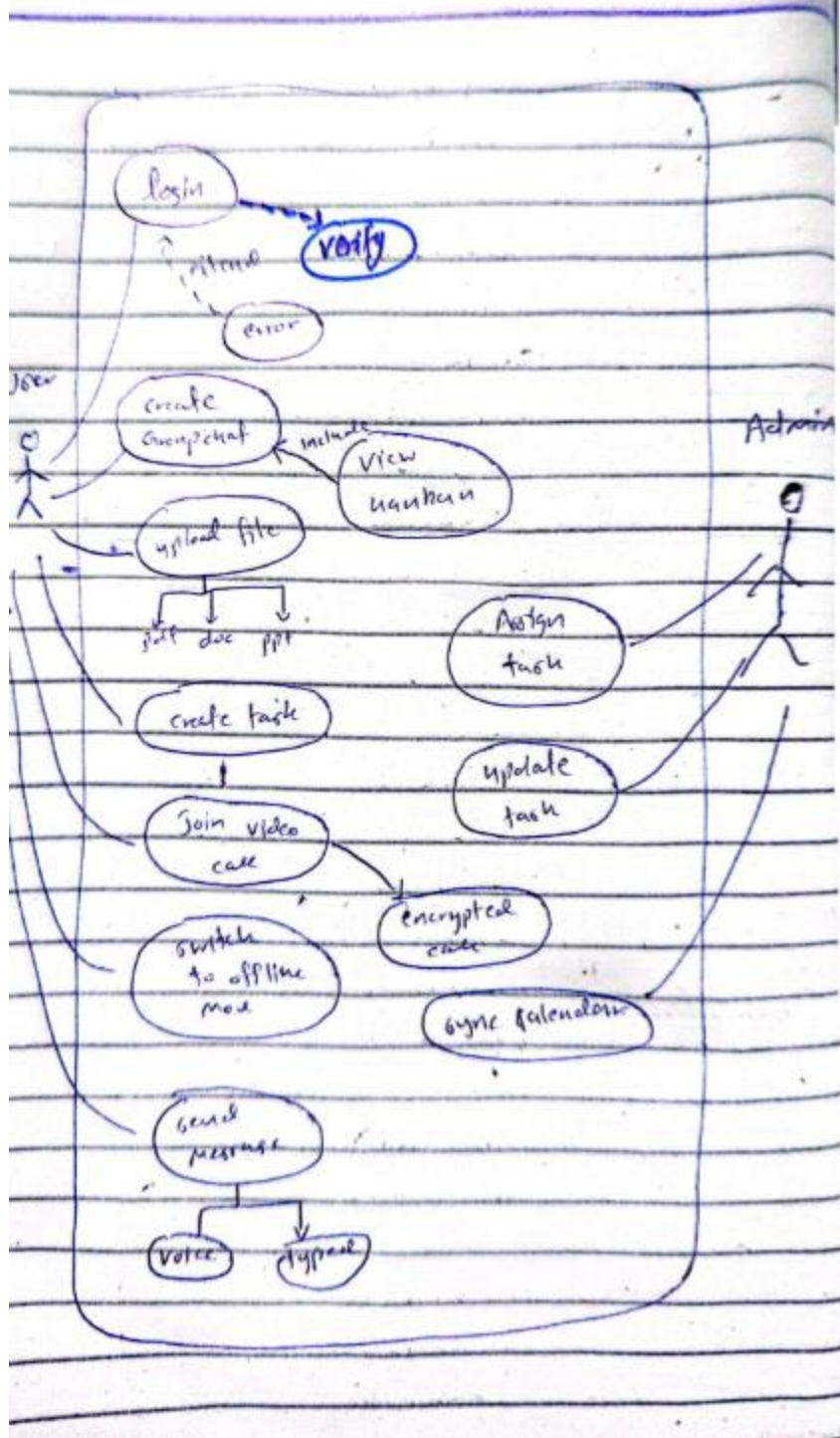
## Sequence Diagrams for Task Assignment, Chat, and Video Conferencing



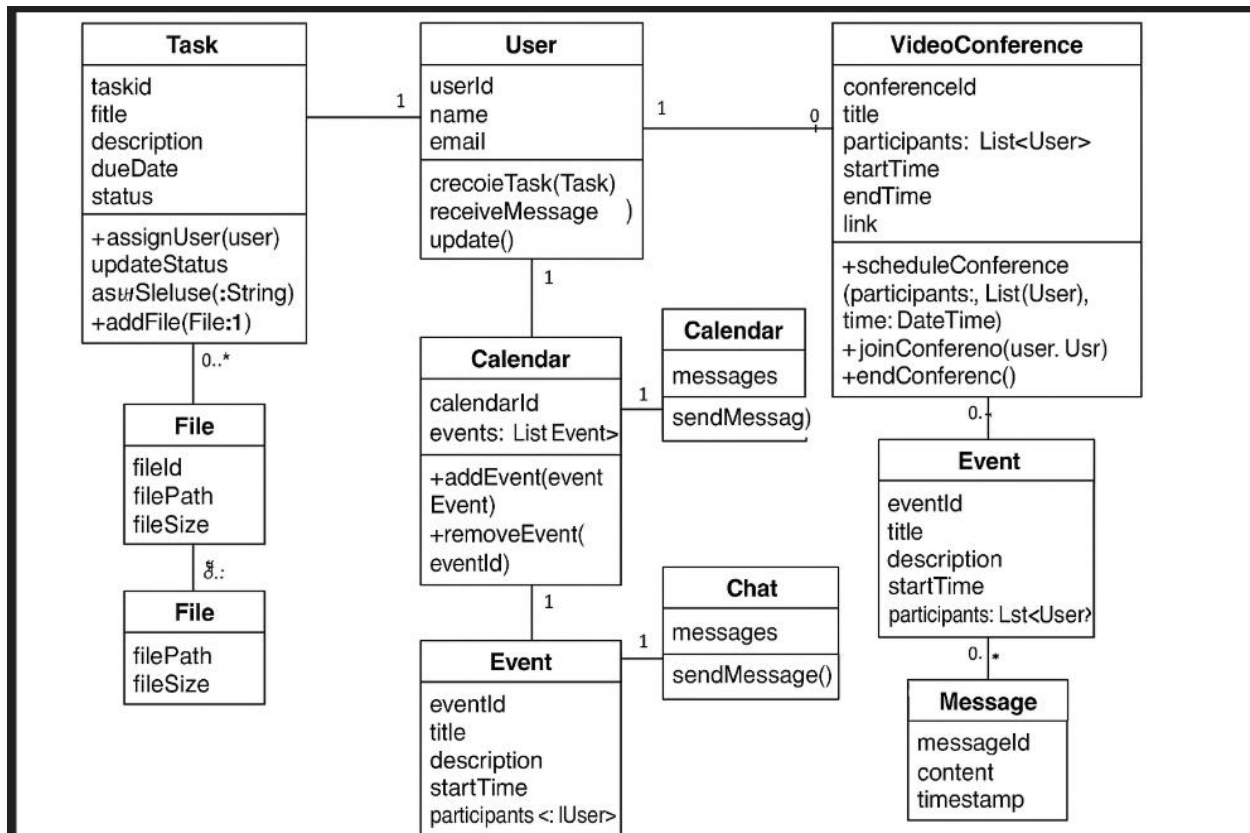




Use Case Diagram



Class Diagram



## d) Architecture

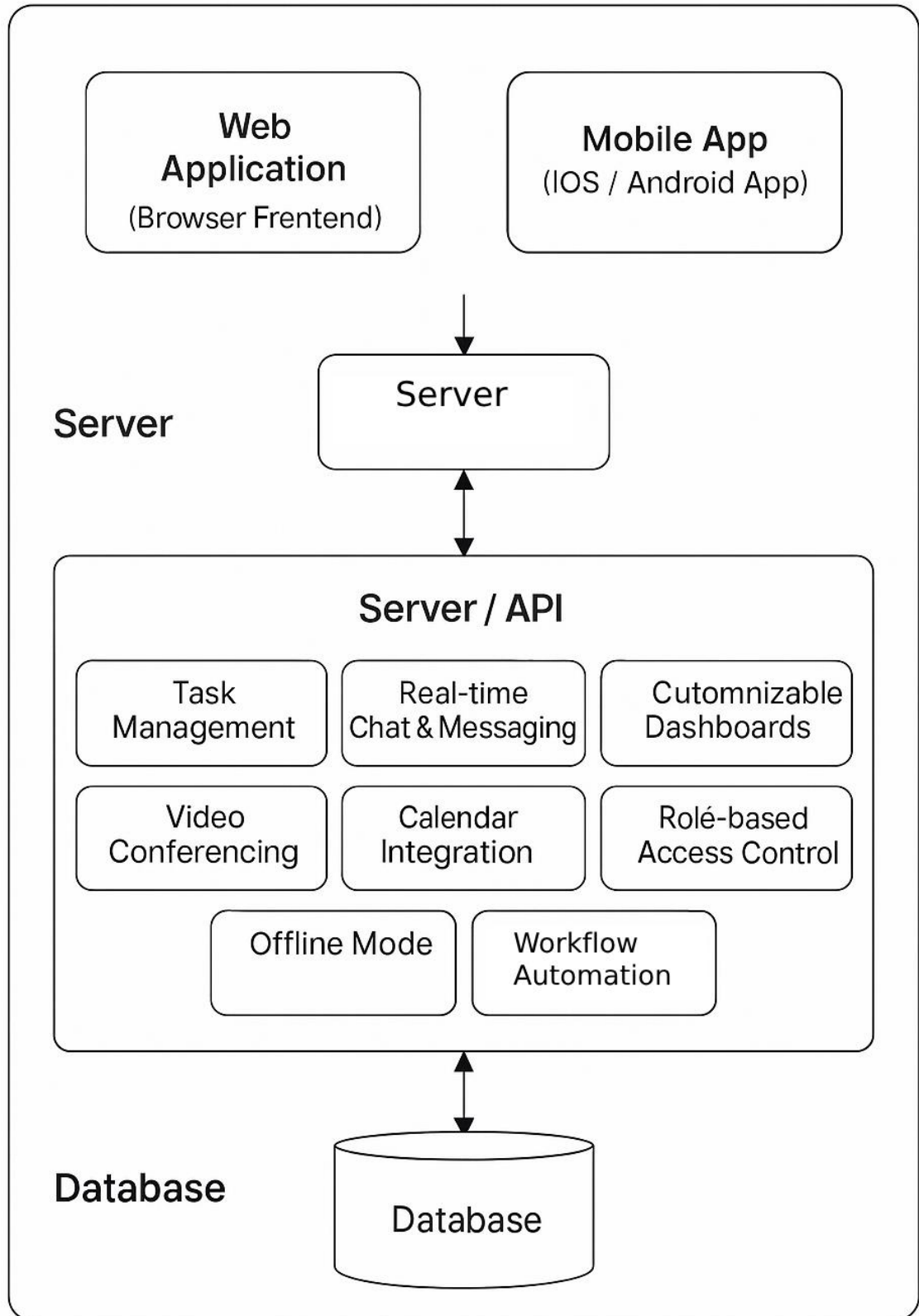
For **WorkSync Hub**, we have selected a **Client-Based Architecture**.

### Why Client-Based Architecture?

The client-based architecture allows for distributed user interfaces across platforms (web, mobile, desktop) while ensuring that most processing and data storage occurs on the server side. This architecture supports:

- Centralized control and security
- Efficient syncing across multiple clients
- Real-time updates and communication
- Scalability through API-based access to functionalities

# WorkSync Hub



## e) Actual Implementation Screenshots

WorkSync Hub - Login

**WorkSync Hub**

Username:

Password:

WorkSync Hub

**farhan**  
Admin

- Dashboard
- ☒ Tasks
- Chat
- Files
- Video Conference
- Calendar
- Workflows
- Role Management

Trigger: TaskCreated | Conditions: Assignee Equals ali | Actions: SendChat  
Trigger: FileUploaded | Conditions: FileType Equals pdf | Actions: NotifyAllUsers

WorkSync Hub

farhan  
Admin

Dashboard

Tasks

Chat

Files

Video Conference

Calendar

Workflows

Role Management

farhan | Role: Admin  
bital | Role: Member  
ali | Role: Manager  
farhanAli | Role: Member  
123 | Role: Member

Edit Role

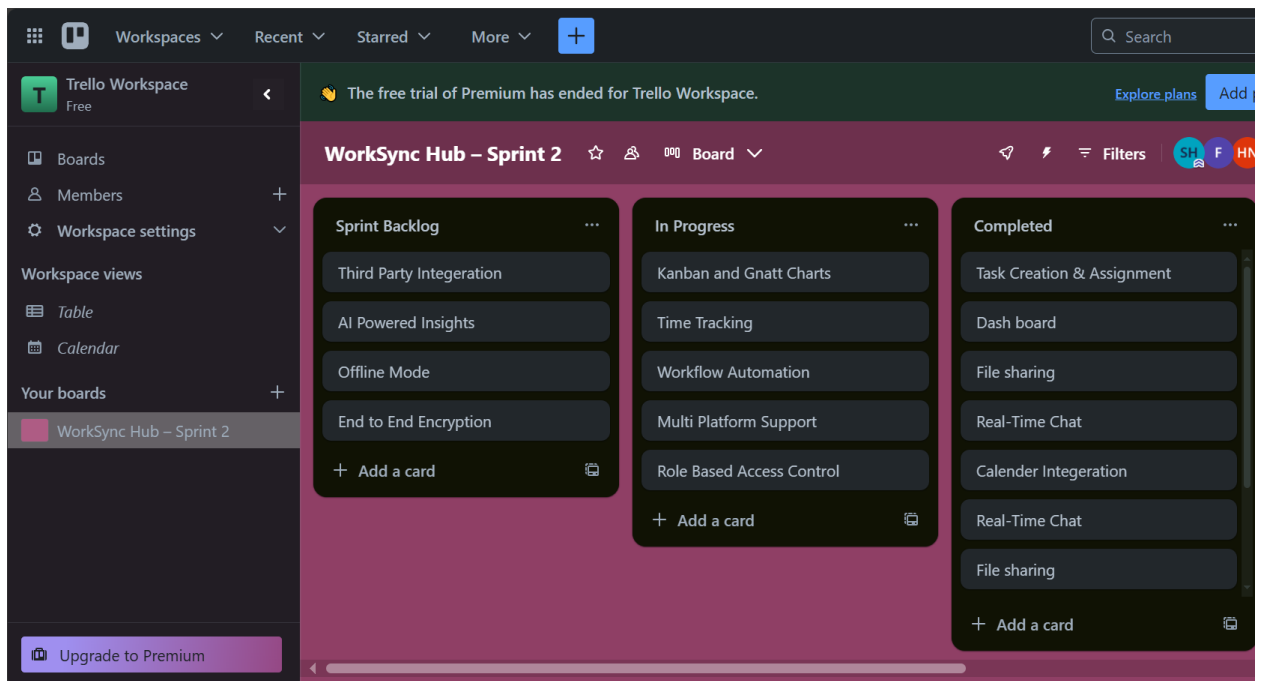
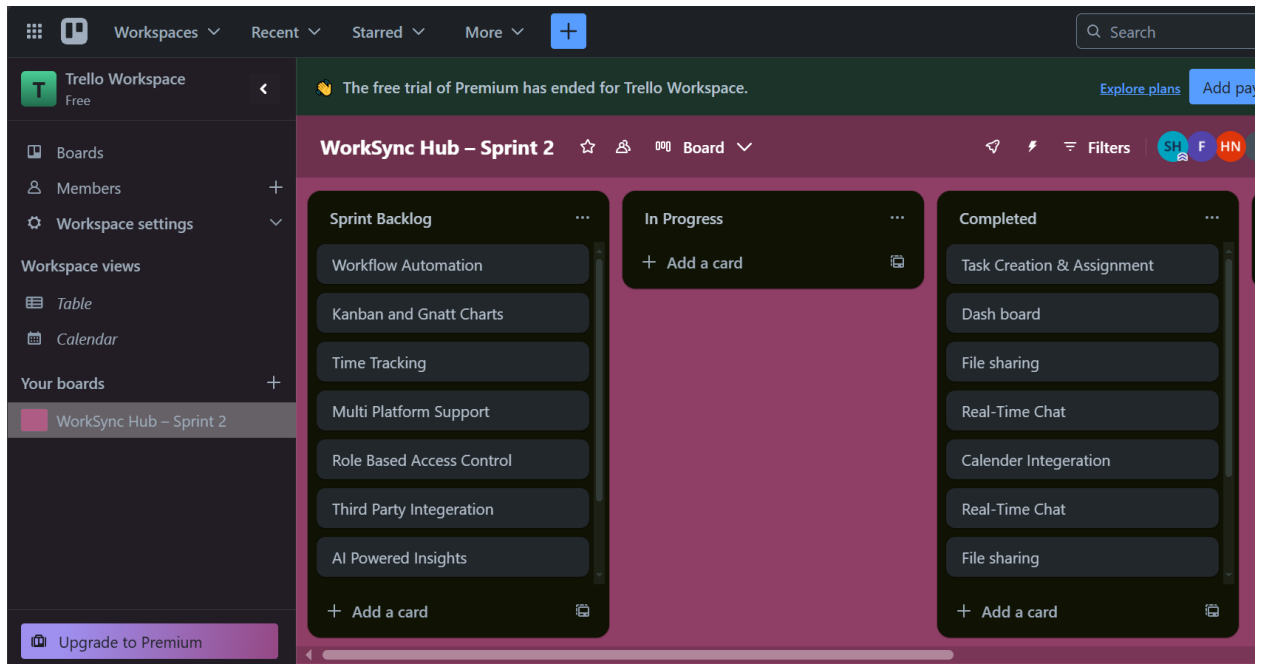
## f) Product Burndown Chart

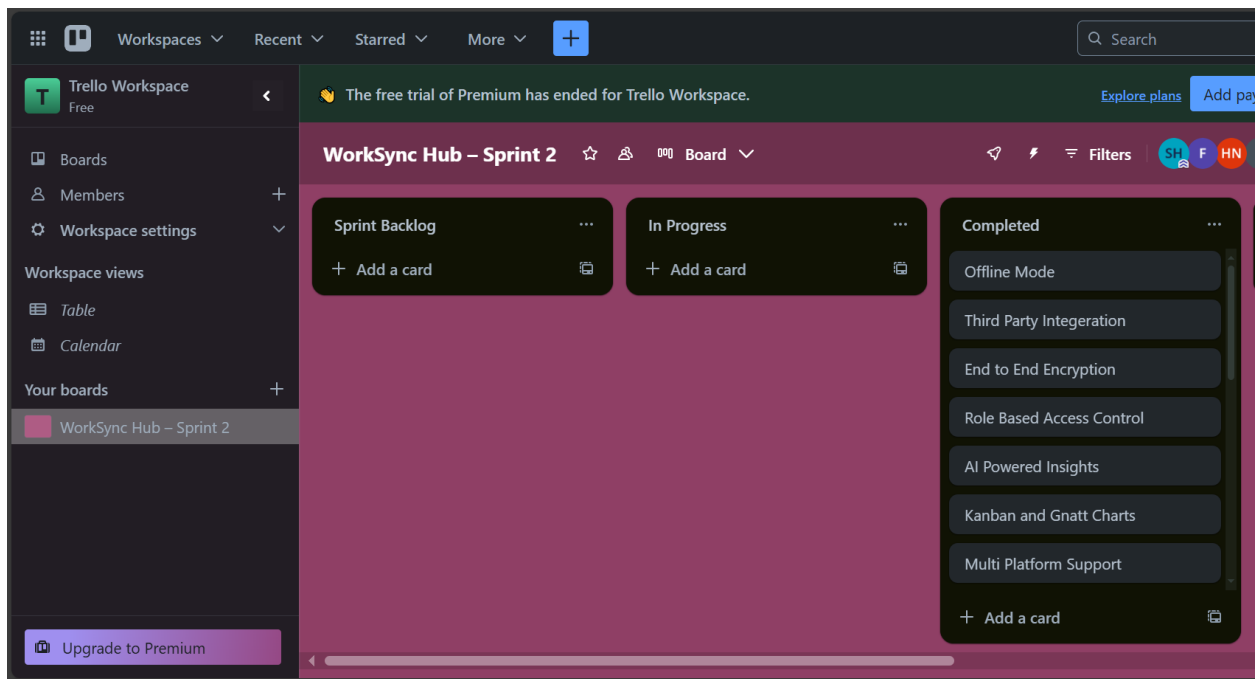
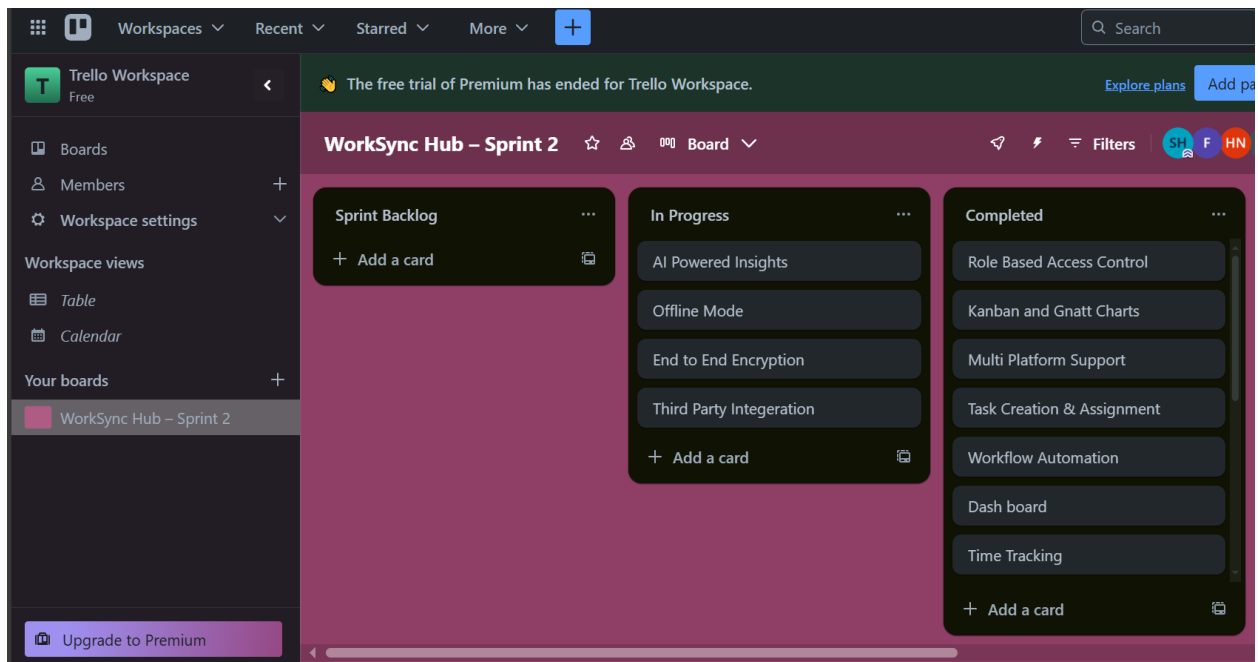
WorkSync Hub - Product Burndown Chart

The chart displays the progress of user stories over a 10-day period. The Y-axis represents 'User Stories Remaining' (0 to 15), and the X-axis represents 'Day' (0 to 10). The Ideal Burndown is a straight dashed blue line from (0, 15) to (10, 0). The Actual Burndown is a solid green line with markers at each day, showing a slightly slower initial pace but staying very close to the ideal line.

| Day | Ideal Burndown | Actual Burndown |
|-----|----------------|-----------------|
| 0   | 15             | 15              |
| 1   | 13.5           | 14              |
| 2   | 12             | 13              |
| 3   | 10.5           | 11              |
| 4   | 9              | 10              |
| 5   | 7.5            | 8               |
| 6   | 6              | 6               |
| 7   | 4.5            | 4               |
| 8   | 3              | 3               |
| 9   | 1.5            | 1               |
| 10  | 0              | 0               |

## g) Trello Board Screenshots





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Here's the **Boundary Value Analysis Testing** section for your report, based on the requirement that *"username and password must match"*:

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## h) Boundary Value Analysis Testing

## Interface Tested: Login and Signup Forms

### Requirement:

- The **username** and **password** must exactly match the stored credentials to allow access.

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### Boundary Value Analysis Table – Login Interface

| Test Case | Username Input | Password Input | Expected Result | Remarks                            |
|-----------|----------------|----------------|-----------------|------------------------------------|
| TC1       | (Blank)        | (Blank)        | Login fails     | Both fields empty – invalid        |
| TC2       | ValidUser      | (Blank)        | Login fails     | Password missing                   |
| TC3       | (Blank)        | ValidPass      | Login fails     | Username missing                   |
| TC4       | ValidUs        | ValidPas       | Login fails     | Slightly shorter than required     |
| TC5       | ValidUser      | ValidPass      | Login success   | Boundary match – exact credentials |
| TC6       | ValidUser1     | ValidPass      | Login fails     | Extra character in username        |
| TC7       | ValidUser      | ValidPass1     | Login fails     | Extra character in password        |

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### Conclusion:

Boundary value analysis confirms that **exact matching** of username and password is mandatory. Even small deviations (missing, extra, or mismatched case) result in a failed login. This ensures secure access based on strict input validation.

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## i) Work Division Between Group Members

| Name          | ID      | Role                  | Responsibilities                                   |
|---------------|---------|-----------------------|----------------------------------------------------|
| Haseeb Nadeem | L232646 | Team Lead & Developer | Developed Real-Time Chat & Task Management Backend |

| Name           | ID      | Role                          | Responsibilities                                        |
|----------------|---------|-------------------------------|---------------------------------------------------------|
| Farhan Ali     | L232530 | Requirement Engineer & Tester | Defined user stories, wrote test cases, tested modules  |
| Shabeeb Haider | P230649 | UI/UX Designer & Developer    | Designed and implemented UI for Chat, Dashboard & Tasks |

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## j) Lessons Learned by Group

- Proper sprint planning and user story writing improves clarity and task management.
  - Integration between different modules requires early planning and API standardization.
  - Communication and collaboration within the team are critical for timely delivery.
  - Testing from the user's perspective helped catch usability issues early.
  - Use of tools like Trello, GitHub, and automated testing enhanced efficiency.
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